

Gesture Control Car Action Group (PTZ)

1. Experiment Objective

Recognizing 5 gestures triggers preset chassis action sequences (square/circle/S-curve/back-and-forth/stop), with an odometry-driven PD closed loop completing the determination. The PTZ pan-tilt switches its angle according to the state.

2. Experiment Principle

1. Key Terms

Term	Description
Action Group	A preset chassis motion sequence; odometry data drives the phase transitions
Odometry Closed-Loop PD	Subscribes to <code>/odom_raw</code> ; each phase drives to the target distance/angle with PD, then cools down before the next phase
Gesture-to-Action Mapping	Yes→square / OK→circle / Rock→S-curve / ThumbDown→back-and-forth / 5 fingers→emergency stop
Gesture Confirmation Count	A gesture must be held for 3 consecutive frames to trigger an action, preventing false triggers
Phase Cooldown Period	After the PD settles, stops for 10 ticks to let the robot stabilize before switching phases
PTZ State Switching	waiting state (0°, -15°); action state (0°, -45°) tilts down to observe the ground
QoS	best_effort (image/odometry + cmd_vel), depth=1

2. Technical Architecture

Three-state machine (idle/waiting/action) + dual timers (image at 15 Hz + action at 30 Hz):

```
/camera/image_raw (BEST_EFFORT)
  → gesture_action_group_control_ptz_node
    └─ Image processing (15 Hz): MediaPipe → gesture recognition → confirmation
count → trigger action
    └─ Action control (30 Hz): odometry PD closed loop → phase transition →
/cmd_vel (BEST_EFFORT)
  └─ PTZ: pan-tilt angle switched by state (waiting: 0°, -15° / action: 0°,
-45°)
/odom_raw (BEST_EFFORT) → odometry-driven PD
```

3. Core Topics Quick Reference

Firmware Downlink

Topic	Message Type	QoS	Field	Range	Direction
/cmd_vel	geometry_msgs/Twist	best_effort + keep_last(1)	linear.x	-0.20~0.20 m/s	$\pm$ forward, $\pm$ backward
			angular.z	-1.00~1.00 rad/s	$\pm$ turn left, $\pm$ turn right
/cmd_ptz_angle	microros_v2_msgs/PtzAngle	reliable	horizontal_deg	0°	Fixed horizontal
			pitch_deg	-45°~15°	Switched by state

3. Experiment Steps

1. Hardware Preparation

Hardware	Requirement
PTZ version	☒
No-PTZ version	☐
Required peripherals	ESP32 camera (pan-tilt) + PTZ servo

2. Activate the MediaPipe Environment

```
source /opt/ros/humble/setup.bash
source ~/microros_ws/install/setup.bash
```

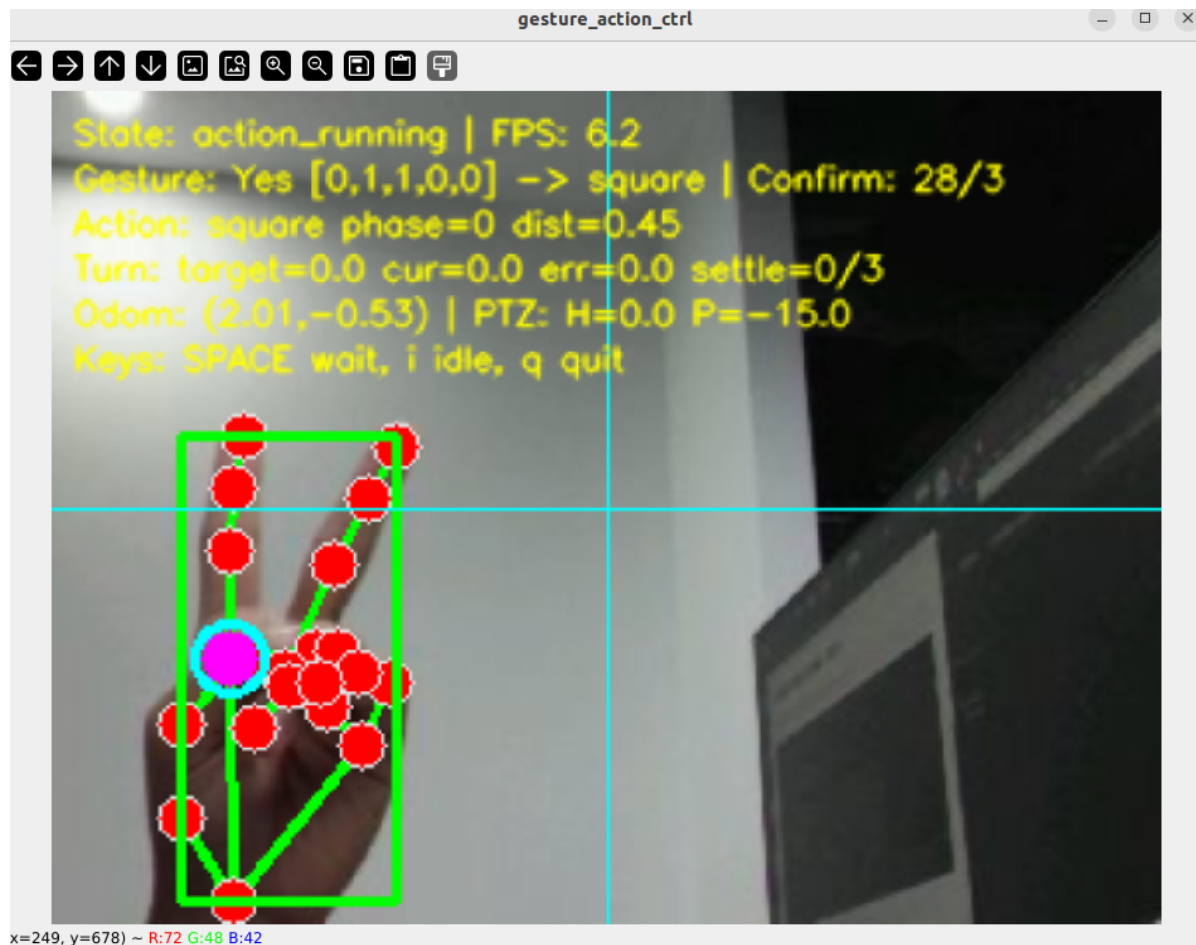
3. Start Agent + Camera + Joint Model (Terminal 1)

```
ros2 launch microros_v2_bringup car_base.launch.py
```

```
yahboom@yahboom:~$ ros2 launch microros_v2_bringup car_base.launch.py
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-07-28-15-04-46-269642-yahboom-579370
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [micro_ros_agent-1]: process started with pid [579411]
[INFO] [camera_driver-2]: process started with pid [579413]
[INFO] [robot_state_publisher-3]: process started with pid [579415]
[INFO] [joint_state_publisher-4]: process started with pid [579417]
[INFO] [ekf_node-5]: process started with pid [579419]
[micro_ros_agent-1] [1785222287.404236] info | UDPv4AgentLinux.cpp | init | running... | port: 8090
[robot_state_publisher-3] [INFO] [1785222288.847816851] [robot_state_publisher]: got segment base_footprint
[robot_state_publisher-3] [INFO] [1785222288.848013042] [robot_state_publisher]: got segment base_link
[robot_state_publisher-3] [INFO] [1785222288.848028942] [robot_state_publisher]: got segment bwheel_Link
[robot_state_publisher-3] [INFO] [1785222288.848037122] [robot_state_publisher]: got segment cam1_Link
[robot_state_publisher-3] [INFO] [1785222288.848043874] [robot_state_publisher]: got segment cam2_Link
[robot_state_publisher-3] [INFO] [1785222288.848050538] [robot_state_publisher]: got segment cam3_Link
[robot_state_publisher-3] [INFO] [1785222288.848057264] [robot_state_publisher]: got segment imu_frame
[robot_state_publisher-3] [INFO] [1785222288.848063747] [robot_state_publisher]: got segment laser_frame
[robot_state_publisher-3] [INFO] [1785222288.848070240] [robot_state_publisher]: got segment lfwheel_Link
[robot_state_publisher-3] [INFO] [1785222288.848076888] [robot_state_publisher]: got segment rfwheel_Link
[joint_state_publisher-4] [INFO] [1785222289.372349515] [joint_state_publisher]: Waiting for robot_description to be published on the robot_description
[camera_driver-2] [INFO] [1785222289.716161193] [esp32_ip_camera_publisher]: camera node started, camera_url: http://192.168.12.37:81/stream
[camera_driver-2] [WARN] [1785222292.705741414] [esp32_ip_camera_publisher]: Camera not opened, trying reconnect... (next retry in 1.0s)
[camera_driver-2] [INFO] [1785222294.072701944] [esp32_ip_camera_publisher]: IP camera stream opened successfully
```

4. Start Gesture Action Group Control (Terminal 2)

```
ros2 launch microros_v2_ptz_visual_control_pkg  
gesture_action_group_control_ptz.launch.py
```



5. Operation Instructions

Gesture-to-Action Mapping

Gesture	Finger Code	Action
Yes	[0,1,1,0,0]	square (0.5 m square, 4 sides)
OK	≥3 fingers, index bent	circle (spin 360° in place)
Rock	[0,1,0,0,1]	s_curve (S curve)
Thumb Down	≤1 finger, thumb down	backforth (back-and-forth 0.5 m)
5 fingers extended	[1,1,1,1,1]	stop (emergency stop)

Key	Function
Space	waiting ↔ idle
i / I	Back to idle
q / Q / ESC	Exit

4. Experiment Observations

- **idle:** PTZ (0°, -15°), gestures are detected but do not trigger actions
- **waiting:** The car body is stationary, waiting for a gesture
- **Yes gesture held for 3 frames:** 0.5 m square (4 sides), PTZ tilts down to -45°
- **OK gesture:** Spins 360° in place
- **Rock gesture:** S curve
- **Thumb Down:** Back-and-forth 0.5 m
- **5-finger emergency stop:** Interrupts immediately and returns to waiting

5. Program Analysis

Source code paths:

- Launch:
`~/microros_ws/src/microros_v2_ptz_visual_control_pkg/launch/gesture_action_group_control_ptz.launch.py`
- Node:
`~/microros_ws/src/microros_v2_ptz_visual_control_pkg/microros_v2_ptz_visual_control_pkg/gesture_action_group_control_ptz_node.py`
- Hand detector:
`~/microros_ws/src/microros_v2_ptz_visual_control_pkg/microros_v2_ptz_visual_control_pkg/hand_detector.py`

1. Core Node Analysis

`gesture_action_group_control_ptz_node` adds PTZ state switching on top of the No-PTZ version: the pan-tilt angle is (0°, -15°) in the waiting state, and switches to (0°, -45°) in the action state to tilt down and observe the ground motion.

PTZ state-switching logic:

```
# Source file:
~/microros_ws/src/microros_v2_ptz_visual_control_pkg/microros_v2_ptz_visual_control_pkg/gesture_action_group_control_ptz_node.py

def _enter_waiting(self) -> None:
    """waiting state: pan-tilt (0°, -15°)"""
    self._state = STATE_WAITING
    self._set_ptz(self._waiting_h_deg, self._waiting_p_deg) # (0°, -15°)
    self._pub_zero_cmd()

def _start_action(self, gesture_key: str) -> None:
    """action state: pan-tilt (0°, -45°) tilted down"""
    self._action_type = gesture_key
    self._set_ptz(0.0, -45.0) # action: tilt down to observe the ground
    self._state = STATE_ACTION
```

Key logic: three-state machine (idle/waiting/action) + odometry PD closed loop + PTZ pan-tilt state linkage. The 5-finger emergency stop can interrupt any action and return to waiting.

2. Key Parameters

Parameter definition files:

- Launch:
`~/microros_ws/src/microros_v2_ptz_visual_control_pkg/launch/gesture_action_group_control_ptz.launch.py`
- Node:
`~/microros_ws/src/microros_v2_ptz_visual_control_pkg/microros_v2_ptz_visual_control_pkg/gesture_action_group_control_ptz_node.py`

Parameter	Type	Default Value	Description
<code>linear_kp</code> / <code>linear_kd</code>	float	<code>0.5</code> / <code>0.1</code>	Distance PD gains
<code>angular_kp</code> / <code>angular_kd</code>	float	<code>0.08</code> / <code>0.15</code>	Angle PD gains
<code>max_linear_x</code>	float	<code>0.2</code> m/s	Maximum linear velocity
<code>max_angular_z</code>	float	<code>1.0</code> rad/s	Maximum angular velocity
<code>distance_tolerance</code>	float	<code>0.03</code> m	Distance determination tolerance
<code>angle_tolerance_deg</code>	float	<code>5.0</code> °	Angle determination tolerance
<code>settle_frames</code>	int	<code>5</code>	Frames for the settle determination
<code>phase_cooldown_ticks</code>	int	<code>10</code>	Cooldown ticks between phases
<code>gesture_hold_confirm_frames</code>	int	<code>3</code>	Gesture confirmation frames
<code>square_forward_distance</code>	float	<code>0.5</code> m	Square side length
<code>circle_total_angle_deg</code>	float	<code>360.0</code> °	Total circle angle
<code>backforth_distance</code>	float	<code>0.5</code> m	Back-and-forth distance
<code>waiting_horizontal_deg</code>	float	<code>0.0</code> °	waiting pan-tilt horizontal angle
<code>waiting_pitch_deg</code>	float	<code>-15.0</code> °	waiting pan-tilt pitch angle

3. Node Description

Node	Package	Description
<code>gesture_action_group_control_ptz_node</code>	<code>microros_v2_ptz_visual_control_pkg</code>	PTZ gesture action group, three-state machine + PD closed loop + pan-tilt linkage

6. Frequently Asked Questions

Problem	Cause	Solution
Action group interrupted	No data on the <code>/odom_raw</code> odometry topic	Check the odometry topic
Gesture does not trigger	Not enough confirmation frames	Hold the gesture for more than 3 frames
Square drifts off course	Angle PD parameters not suited to the floor	Tune <code>angular_kp</code> / <code>angular_kd</code>
No detection at startup	The MediaPipe environment was not sourced	<pre>source /opt/ros/humble/setup.bash source ~/microros_ws/install/setup.bash</pre>